Heat (p99-105)

Heat is a form of energy that you can sense through receptors in your skin. Heat can be lost from your skin when you stand in front of a fan or be gained when your body absorbs radiant heat from a log fire.

Heat

1. What is heat?

2. Explain why a saucepan with 2 litres of water at 100°C has more heat energy than a saucepan with only 1 litre of water at the same temperature. Use a diagram to explain your answer.

3. Why does a small amount of water heat up more than a large amount of water when the same amount of energy is supplied? Use a diagram to help your answer.

Temperature

1. What is temperature?

2. What is commonly used to measure temperature and how do they work?

3. Name the three scales used to measure temperature.

Heat Transfer

1. In which direction does heat flow?

2. What are the three ways heat transfer can happen?
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Conduction

1. How can heat be conducted through an object? (hint: look at figure 4.1.6)

2. What are conductors? Give an example of a good conductor of heat.

3. What are insulators? Give three examples of insulators.

4. Why does trapped air in woollen jumpers and blankets keep us warm? What other objects also keep us warm in this way?

5. Name adaptations polar bears and penguins have to keep warm.

Convection

1. Explain how heat is transferred through convection.

2. Why can heat be transferred through convection in liquids and gases but not solids?

3. Use figure 4.1.11 to explain how water heats by convection. Draw a diagram as part of your answer.

Radiation

1. Why can't heat from the sun travel by conduction or convection?

2. How does radiation transmit heat?

3. The hotter an object is, the _____ heat it radiates.

4. When radiated energy hits a surface, the heat can be absorbed, reflected or transmitted. Describe the types of materials which will:
    a. Absorb heat

    b. Reflect heat

c. Transmit heat

5. Explain how solar hot water systems use heat absorption from the sun to heat water. Use a diagram in your answer.

Temperature Scales (p101)

There are three commonly used temperature scales. Describe each scale, including who developed it and when, important numerical values on the scale and where it is used.

Fahrenheit

Celsius

Kelvin